Sindh Science Learning Portal

Class 5 · Science

Chapter 6

Force and Simple Machines



Learn about pushes and pulls, what forces do, friction, and six simple machines that make work easier.

Following the Sindh Textbook Board curriculum

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CHAPTER 6

Force and Simple Machines

What we will learn Learn about pushes and pulls, what forces do, friction, and six simple machines that make work easier.

1 What is a force?

A **push** or a **pull** is called a **force**. To open a door we push it or pull it.

A force can:

- **start** something moving, like kicking a football.
- **stop** something that is moving, like catching a ball.
- change the **direction** or **speed** of something, like hitting a cricket ball with a bat.
- change the **shape** of something, like making a cup out of clay.

2 Some kinds of force

- **Muscular force**: the strength of our body, like lifting a bag or riding a bicycle.
- **Gravity**: the Earth pulls everything down towards it. That is why a ball thrown up comes back down.
- **Friction**: when two surfaces rub together, friction slows a moving thing down. There is more friction on rough surfaces and less on smooth ones. Friction is also what lets us walk without slipping.
- **Magnetic force**: a magnet pulls iron towards it, even without touching it.

3 Simple machines

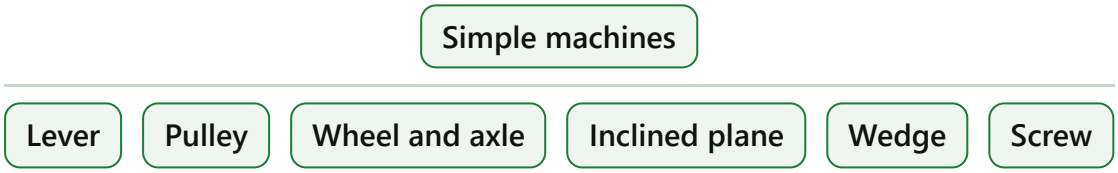
Simple machines are tools that make work **easier** using less force. There are six simple machines:

- **Lever**: a strong bar that turns on one point. Examples: a see-saw, a spoon used to open a tin lid.
- **Pulley**: a wheel with a rope running over it. Examples: pulling water from a well, raising a flag.
- **Wheel and axle**: a wheel that turns together with a rod (axle). Examples: a bicycle, a cart, a door handle.
- **Inclined plane (ramp)**: a sloping surface that makes it easier to move heavy things up. Example: a plank for loading a truck.
- **Wedge**: used to cut or split things apart. Examples: an axe, a knife.
- **Screw**: holds things together when it is turned. Examples: a bottle cap, a screw.



Picture: Killarnee · CC BY-SA 4.0 · Wikimedia Commons

Diagram: the parts and what they do



Simple machines (سادا مشينون · سادہ مشینیں)
Tools that make work easier with less force.

Lever (ليور · لیور)
A bar that turns on one point. Examples: see-saw, opening a lid with a spoon.

Pulley (پُلي · پُلی)
A wheel and rope that make lifting easier. Examples: a well, a flagpole.

Wheel and axle (ڦيٽو ۽ ڏُرو · پہیہ اور دُھرا)
A wheel and rod that turn together. Examples: bicycle, cart.

Inclined plane (لاهي واري سطح · ڈھلوان سطح)
A slope that makes moving heavy things up easier. Example: a ramp.

Wedge (ويج · فانہ)
A sharp edge that splits things. Examples: axe, knife.

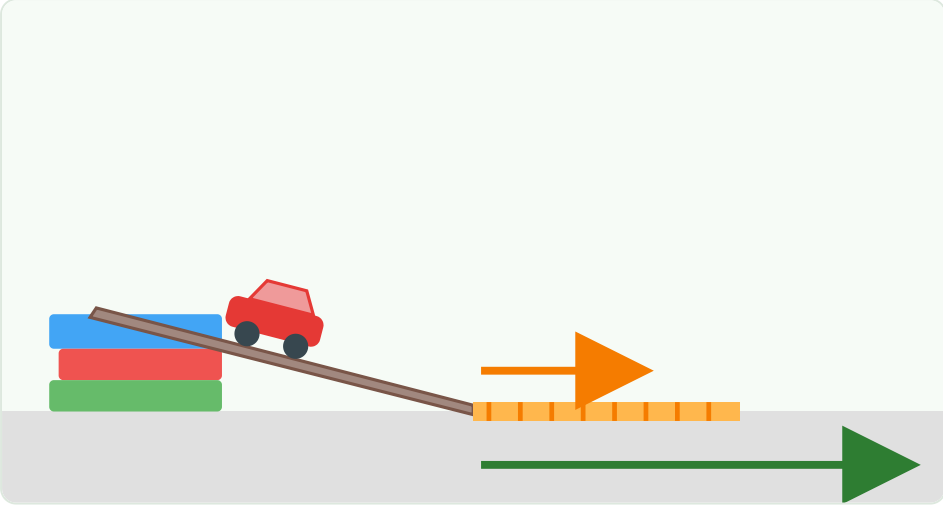
Screw (پيچ · پیچ)
Holds things tightly when turned. Example: a bottle cap.

Key words

English	سنڌي	اردو
force	قوت	قوت
push	ڌڪو	دھکا

English	سنڌي	اردو
pull	ڇڪ	کھنچاؤ
gravity	ڪشش ثقل	کششِ ثقل
friction	رگڑ	رگڑ
simple machine	سادي مشين	سادہ مشین
lever	ليور	لیور
pulley	پُلي	پُلی
wheel and axle	ڦيٽو ۽ ڏُرو	پہیہ اور دُھرا
inclined plane	لاهي واري سطح	ڈھلوان سطح

Experiment: Friction on rough and smooth surfaces



You need

- A toy car or a small ball
- A book and a board or thick card
- A towel or a rug

Steps

1. Lean the board on the book to make a ramp.
2. Let the car roll down and see how far it goes on the smooth floor.
3. Now spread the towel at the bottom of the ramp and let the car roll again.
4. Compare how far the car went each time.

Safety: Do the experiment on the floor and never throw the car at anyone.

What did we learn? The car stops sooner on the towel. The rough towel makes more friction, which slows the car down. The smooth floor has less friction, so the car goes further.

Practice questions

Circle the correct answer

1. **What is a force?**

   (a) A push or a pull
   (b) A colour
   (c) A sound
   (d) A plant

2. **When we make a cup from clay, what does the force change?**

   (a) The colour
   (b) The shape
   (c) The mass
   (d) The taste

3. **Why does a ball thrown up come back down?**

   (a) Because of friction
   (b) Because of gravity
   (c) Because of the wind
   (d) Because of light

4. **On which surface is there the most friction?**

   (a) Ice
   (b) Smooth glass
   (c) A rough stony road
   (d) An oily floor

5. **A see-saw is which simple machine?**

   (a) Lever
   (b) Screw
   (c) Wedge
   (d) Pulley

6. **Which simple machine is used to pull water from a well?**

   (a) Wedge
   (b) Pulley
   (c) Inclined plane
   (d) Screw

7. **An axe is which simple machine?**

   (a) Wheel and axle
   (b) Pulley
   (c) Wedge
   (d) Lever

8. **A plank used to load heavy things onto a truck is which simple machine?**

   (a) Inclined plane
   (b) Screw
   (c) Pulley
   (d) Wedge

9. **Which force do we use to lift a school bag?**

   (a) Magnetic force
   (b) Muscular force
   (c) Friction
   (d) Light

Write true or false

10. **A force can stop a moving object.** _____

11. **Friction makes moving things go faster.** _____

12. **A magnet can pull iron without touching it.** _____

13. **Simple machines make work harder.** _____

14. **A bicycle has a wheel and axle.** _____

Match the columns

15. Match each simple machine with an example.

1. Lever		(a) Raising a flag
2. Pulley		(b) Knife
3. Wedge		(c) Bottle cap
4. Screw		(d) See-saw

Match the columns

16. Match each action with what the force does.

1. Kicking a football		(a) Stops it
2. Catching a ball		(b) Changes its shape
3. Making a clay cup		(c) Starts it moving

Worksheet: Force and Simple Machines

Name: _____ Date: _____

A. Fill in the blanks

Word box: force · friction · gravity · lever · pulley

1. A push or a pull is called a _____ .
2. The Earth pulls everything down because of _____ .
3. There is more _____ on a rough surface.
4. A _____ is used to pull water from a well.
5. A see-saw is a _____ .

B. Put ✓ for true and ✗ for false

1. ☐ A force can change the direction of an object.
2. ☐ There is a lot of friction on ice.
3. ☐ A knife is a wedge.
4. ☐ Simple machines need more force to do work.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Wheel and axle	( ) Axe
2. Screw	( ) Ramp
3. Wedge	( ) Cart
4. Inclined plane	( ) Bottle cap

D. Answer in short

1. Write two things a force can do.

_____

_____

2. How does friction help us?

_____

_____

3. Name the six simple machines.

_____

_____

E. Draw and label

Draw two simple machines used at your home or school and write their names.

FOR TEACHERS

Answer key

Chapter 6: Force and Simple Machines

Practice questions

1. (a) A push or a pull
2. (b) The shape
3. (b) Because of gravity
4. (c) A rough stony road
5. (a) Lever
6. (b) Pulley
7. (c) Wedge
8. (a) Inclined plane
9. (b) Muscular force
10. True
11. False
12. True
13. False
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. force · 2. gravity · 3. friction · 4. pulley · 5. lever

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Axe · (4) Ramp · (1) Cart · (2) Bottle cap

D. Answer in short:

1. A force can start an object moving and can change its shape.
2. Friction lets us walk without slipping.
3. Lever, pulley, wheel and axle, inclined plane, wedge and screw.

Picture credits

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